

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> For part (a), we use the formula : $q = m * c * \Delta T$.

- m (mass) = 1000 g (1L of water),

- c (specific heat) = 1 cal/gÂ°C ,

- ΔT (change in temperature) = 60Â°C - 30Â°C = 30Â°C .

$$q = 1000 * 1 * 30 = 30000 \text{ cal} = 30 \text{ Kcal (since 1 Kcal = 1000 cal)}.$$

For part (b), similarly , we use the same formula .

- m = 1000 g (1kg of aluminum),

- c = 0.215 cal/gÂ°C ,

- ΔT = 60Â°C - 30Â°C = 30Â°C .

$$q = 1000 * 0.215 * 30 = 6450 \text{ cal} = 6.45 \text{ Kcal (since 1 Kcal = 1000 cal)}.$$

(a) 30 Kcal , (b) 6.45 Kcal .

</think>

<answer> 30 Kcal , 6.45 Kcal </answer>